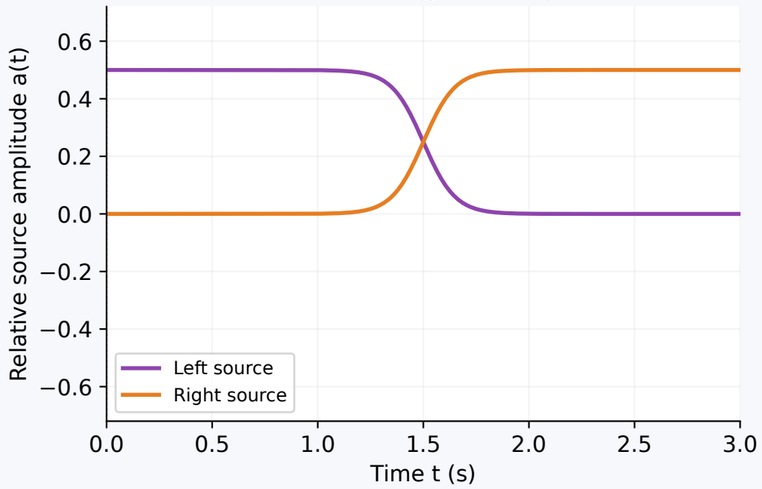
MAGNETIC INPUT → FLUID FLOW → PARTICLE RESPONSE

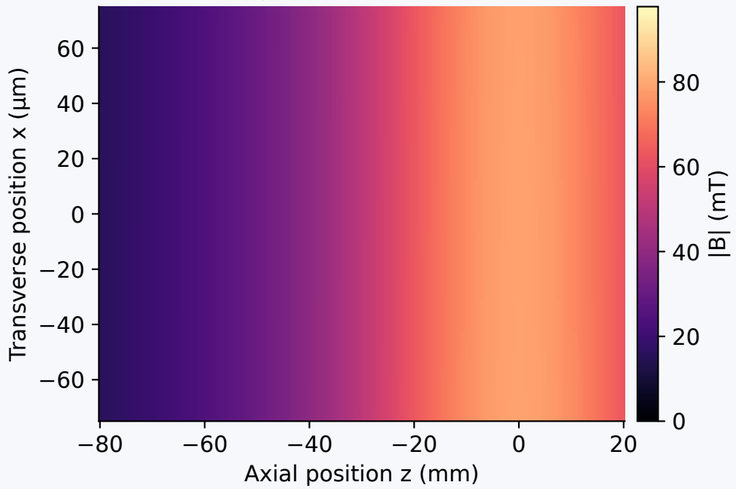
Smooth switch: left to right

$a_{L,R} = 0.25[1 \mp \tanh((t - 1.5)/0.15)]$

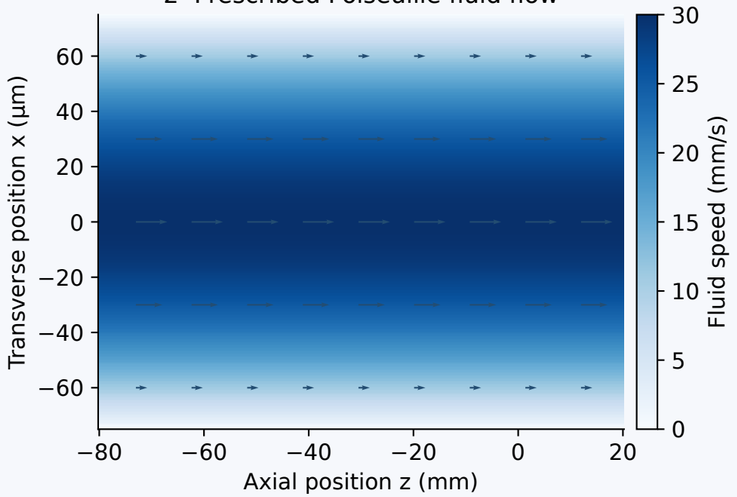
1a Prescribed magnetic amplitudes



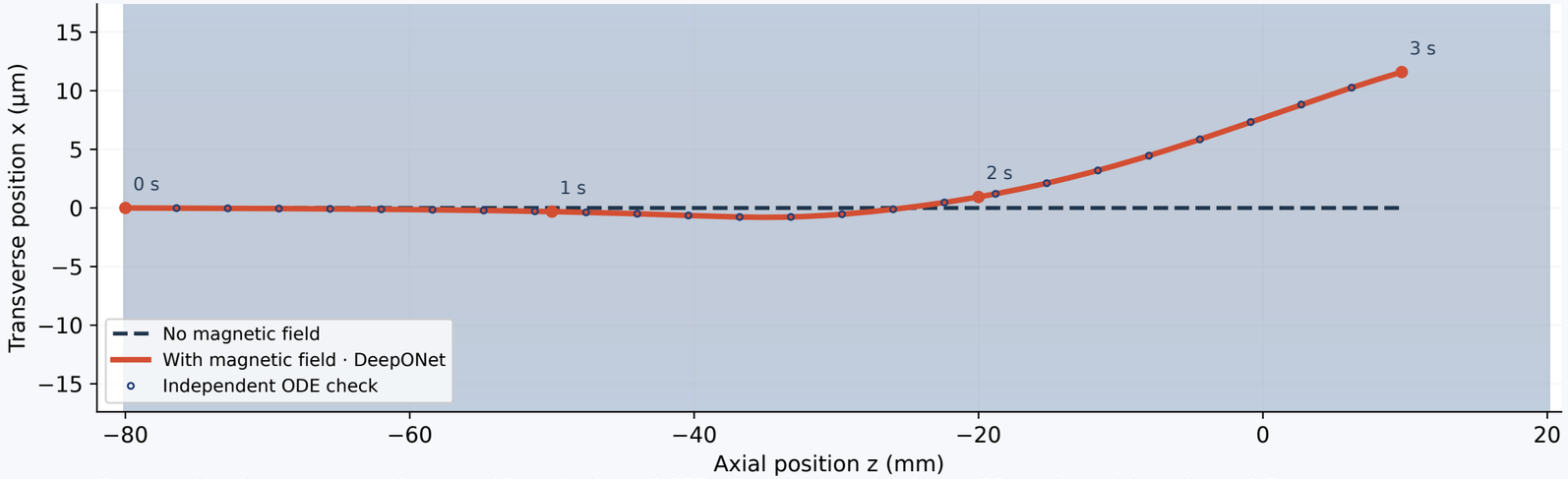
1b Magnetic field at t = 0.00 s



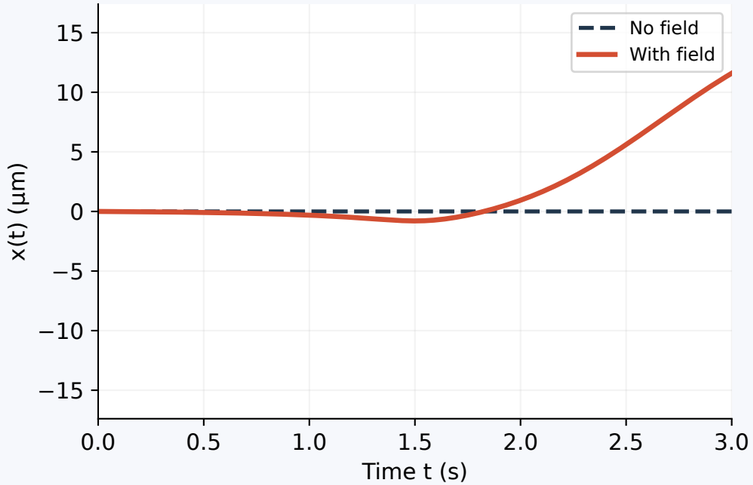
2 Prescribed Poiseuille fluid flow



3a Same release and flow: trajectories with / without magnetic forcing



3b Transverse particle displacement



Same settings in every case: Umean = 15 mm/s | η = 3.277 mPa·s | channel radius = 75 μm | particle radius = 0.5 μm

Time-varying amplitudes; fixed spatial source geometry. Overdamped induced-dipole model. Axial and transverse axes use different units.

Independent max error: x 0.004 μm · z 0.88 μm